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SPIRE Universal Ingest Service

Companion to the SPIRE Commander Brief

BLUF. The Universal Ingest Service (UIS) is the data intake layer underneath SPIRE: 8 file formats, 6 source channel types, 4 pre-built Marine-Corps logistics adapters, and a local LLM that proposes column mappings for everything else. It is air-gap-deployable, audit-chained at the apply step, and extractable from SPIRE as a standalone Python package. None of the surveyed commercial or federal products combine all four of those properties.

What it does

- **8 formats:** CSV, TSV, XLSX, JSON, JSONL, XML, fixed-width, EDI X12.
- **6 channels:** drop, SFTP, IMAP, HTTP poll, DB CDC, Kafka. Each carries retry / backoff, circuit breaker, byte-bounded caps, path-prefix containment, and a dead-letter queue.
- **4 adapters:** GCSS-MC ECP / UTIL / SR-Header, DRRS-MC C-Rating — declarative `AdapterSpec` with types, sensitivity, keys, and constraints. Adding a new adapter is one Python file.
- **Local LLM column mapper.** A 26B Gemma 4 on the same 2U box proposes a mapping for non-canonical CSVs; sample rows are PII-sanitised first; the operator confirms; the mapping persists as a reusable per-unit profile.
- **Audit-chained apply.** Every apply commits a SHA-256 hash-chained, Ed25519-signed audit entry; SIEM export in CEF for Splunk / ArcSight / QRadar.

What is distinctive

A 2026 survey of commercial ETL stacks (Informatica, Talend, Airbyte, Fivetran), federal-targeted ingest (Palantir Foundry, ADVANA, Maven Smart System, CDAO Edge Data Mesh), and OSS (Apache NiFi, Singer, Meltano) found three SPIRE-specific differentiators:

- **Marine-Corps canonical adapters as ship-with-the-product defaults.** GCSS-MC ECP / UTIL / SR-Header and DRRS-MC C-Rating arrive pre-mapped. No commercial or federal product surveyed ships these; Foundry would build them as forward-deployed-engineer work; NiFi as customer flow-files.
- **Tamper-evident audit chain fused with the ingest path.** Hash-chained + Ed25519-signed apply commits, end-to-end. NiFi has tamper-*resistant* provenance, not chained; Foundry has Apollo audit, not chain-on-apply; commercial ETL has audit logs, not cryptographic chains.
- **Operator-tier (not admin-tier) schema mapping via a local LLM.** A unit S-4 drops a strange XLSX; the on-box model proposes the map; the operator confirms; it saves for next time. The combination of on-box LLM, PII sanitisation before the model sees a sample, operator targeting, and per-unit profile reuse is not present in any product surveyed. Foundry AIP is closest; admin-tier and cloud-anchored.

Honest framing

The format × channel matrix is commodity (NiFi has covered it for a decade); air-gap ETL is commodity (NiFi, Airbyte OSS, Foundry on-prem, Informatica PowerCenter); LLM-assisted column mapping in general is commodity (Snowflake, Atlan, Informatica CLAIRE, Foundry AIP). UIS does *not* replace Foundry, ADVANA, or CDAO Edge Data Mesh. It *complements* them: UIS is the unit-side last-mile layer that turns operator-dropped artifacts into canonical rows; higher-tier platforms consume the rows.

Extractability from SPIRE

UIS is built as a standalone Python package (`backend/uis/`) with dependency injection at every coupling point to the rest of SPIRE (`set_audit_func`, `set_connection_factory`). Two soft imports remain into the SPIRE audit-log helper (`verify_chain`, `recent_entries`) both are read-only utilities, easy to inline or inject. Modularisation was a deliberate engineering goal (UIS-24), not an accident. Estimated effort to ship UIS as a separate `pip`-installable library — vendor the package, inline the two helpers or wire to the consumer's audit log, install transitive deps (paramiko, kafka-python, openpyxl, charset_normalizer, cryptography), publish wheel: **1 engineer-day**.

Maturity

TRL 4 inside SPIRE; validated against synthetic GCSS-MC ECP / UTIL / SR-Header at MDM 2026. HTTP, DB CDC, and Kafka channels are protocol- complete extension points pending production hardening. No ATO.

Path forward

- **A sanitised real GCSS-MC export** from a sponsoring MEF G-4 or MLR S-4. The synthetic dataset proves shape; only a real pull proves edge cases.
- **An MCSC POC** to scope UIS as either a SPIRE-internal component or a separately licensable Marine-Corps logistics ingest library — architecture supports either.

POC: Jesse Morgan · Thornveil · jesse@thornveil.ai
Code: github.com/jeranaias/spire/tree/master/backend/uis (public)

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